

Collections Copilot

Evidence-backed assistance for collections employees

Team 5C2 | CIBC Hackathon Build Phase

Local classifier and language generation | Employee review

The employee workflow

Find a trustworthy customer view across five source systems before discussing the case.

Identify financial hardship and disputes during the conversation, then retrieve the applicable policy.

Review a suggested response, record the employee decision, and preserve the evidence.

Measure preparation time against manual lookup. The original 30% improvement target remains unmeasured.

Data foundation and coverage

1,011,944

accepted golden customers

1,142,071 matched accounts | 338,295 served cases

13 foundation checks cover keys, customer ownership, transcript links, amounts and contract schema. Contradictory records are quarantined.

C360 export: 8.3 MB Parquet, with checksum and explicit coverage.

Four connected layers

- 1 Data product: raw sources, validated identities, curated records and golden C360.
- 2 Insight: supported questions run trusted SQL or retrieve current policy sources; unsupported requests are refused.
- 3 Features: 12 versioned case features, shared text extraction and the same classifier tokenizer for training and inference.
- 4 Assistance: conversation help and decision proposals, followed by employee accept, edit or reject.

Assistance during a conversation

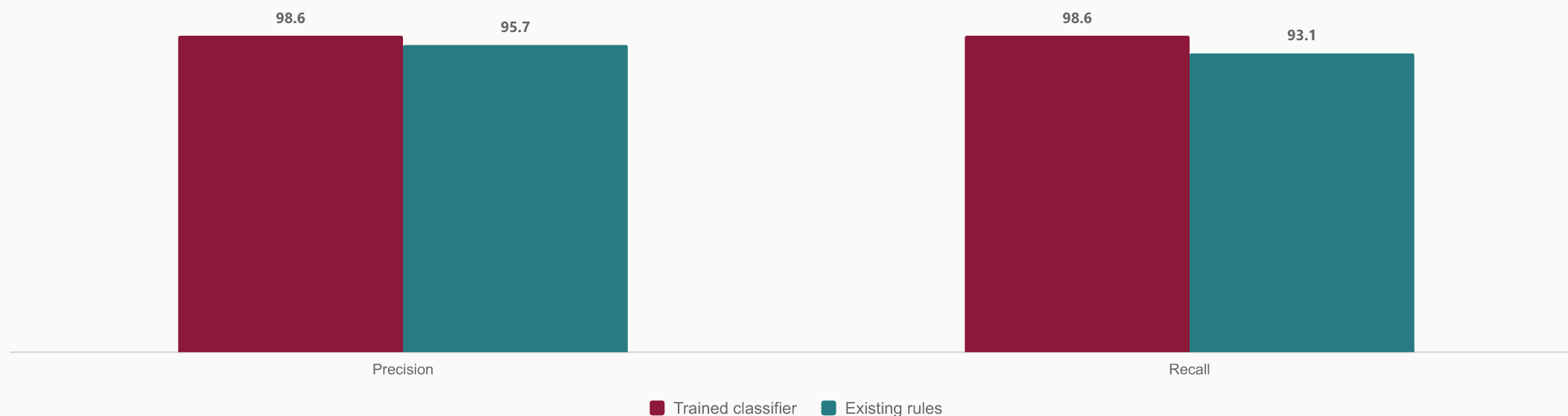
Illustrative customer statement

“I lost my job and cannot afford this payment.”

Acknowledge the hardship. Review support options and affordability before requesting a payment.

Identity verification precedes disclosure. Policy sources and detected signals accompany the draft. Employee review records no automatic action.

Locally trained hardship detection



678 training examples | 181 development evaluation examples

Live combination: 94.7% precision, 100% recall. Public development data only; all flags require review.

A locally fine-tuned language model

Qwen2.5-0.5B-Instruct adapted locally with LoRA on an RTX 4060. 5,40,672 trainable adapter parameters.

1529 factual training examples. Customers remain separate across training, development and final testing.

Final factual-format checks: 468 outputs across 117 cases, including missing values and unsafe instruction challenges.

Complete exact-value output: 100.0%. Contact permission and recommended actions still come from deterministic rules.

Additional reviewed use cases

Next Best Action explains safeguards, alternatives, programme checks and policy conflicts.

Channels show uncertainty and capacity limits. Routing checks skills, French site, support certification and load.

Structured call notes separate customer proposals, explicit commitments, issue mentions and follow-ups with timestamps.

All ten QA items show context and uncertainty. Employees record per-item evidence; no automatic compliance verdict is issued.

Measured checks and remaining work

114 automated checks pass, including the local UI. Public numeric benchmarks pass; 24 policy/scope acceptance checks pass.

Whisper small.en: 11.02% word error rate on 1803 approved synthetic recordings. Speaker attribution still requires review.

CPU Qwen drafts: 10/10 passed typed exact-value checks. Median latency 7.0 seconds; summary usefulness requires employee review.

Next: incorporate the final rubric and extra questions, connect the private repository, publish the C360 link, and record the demo.